

Both Gibbs (G) and Helmholtz (F) free energies are related to other thermodynamic potentials like internal energy (U), enthalpy (H) and entropy (S). They can be interconverted using the relationships: $G = H - TS$, $F = U - TS$, and $H = U + PV$ (where P is pressure and V is volume).

$$\text{So, } G = H - TS = U + PV - TS = PV + (U - TS) = PV + F$$

In ASE, F_vib variable in our code, calls thermo.get_helmholtz_energy(T) function which returns U - TS (U = Internal energy, vibrational contributions only, including zero-point energy,

https://wiki.fysik.dtu.dk/ase/modules/ase/thermochemistry.html#HarmonicThermo.get_helmholtz_energy).

If the adsorbate is bound to the surface, it is essentially a part of the slab. So we can ignore PV and assume $G \approx F$.

ΔG_{ads} : Change in Gibbs free energy when an adsorbate atom adsorbs onto the slab at T starting from a gas phase. ΔG_{ads} determines whether adsorption is thermodynamically favorable ($\Delta G_{\text{ads}} < 0$) or unfavorable ($\Delta G_{\text{ads}} > 0$) at a given temperature (T).

$$\Delta G_{\text{ads}} = G_{\text{slab+adsorbate}} - G_{\text{slab}} - \mu_{\text{adsorbate}}(P)$$

$G_{\text{slab+adsorbate}} = H_{\text{slab+adsorbate}} = E_{\text{slab+adsorbate}} + F_{\text{vib}}$ (NOTE: 1. PV term ignored as mentioned above 2. The Gibbs free energy (G) always includes both electronic and vibrational energies in its internal energy (U) term. But ASE only computes the vibrational part, so we add it back here.)

$G_{\text{slab}} = E_{\text{slab}}$ because we are using frozen slab

$\mu_{\text{adsorbate}}(P) = \mu_0(T) + kT \ln(P/P_0) = E_{\text{adsorbate}} + F_{\text{vib,adsorbate}} + k_B T + k_B T \ln(P/P_0)$ (NOTE: (1) When the adsorbate is free, it behaves as an ideal gas, and it does PV work against external pressure. So, we have to add the $PV = nRT$ term while calculating total energy. Now for 1 mole $PV = RT$ and 1 mole contains N_A (Avogadro number of atoms). So, for a single atom (or single molecule) $PV = (R/N_A)T = k_B T$ where k_B is Boltzmann constant (2) Assume $P = P_0 = 1$, (3) no vibrations for gas, ignore $F_{\text{vib,adsorbate}}$)

Hence we get:

$$\Delta G_{\text{ads}} = (E_{\text{slab+adsorbate}} + F_{\text{vib}}) - E_{\text{slab}} - E_{\text{adsorbate}} - k_B T$$

$$\Delta G_{\text{ads}} = (E_{\text{slab+adsorbate}} - E_{\text{slab}} - E_{\text{adsorbate}}) + F_{\text{vib}} - k_B T$$

$$\Delta G_{\text{ads}} = E_{\text{dft_adsorption_energy}} + F_{\text{vib}} - k_B T$$

Note: adjust to $nk_B T$ if n number of atoms/ molecules in gas phase.